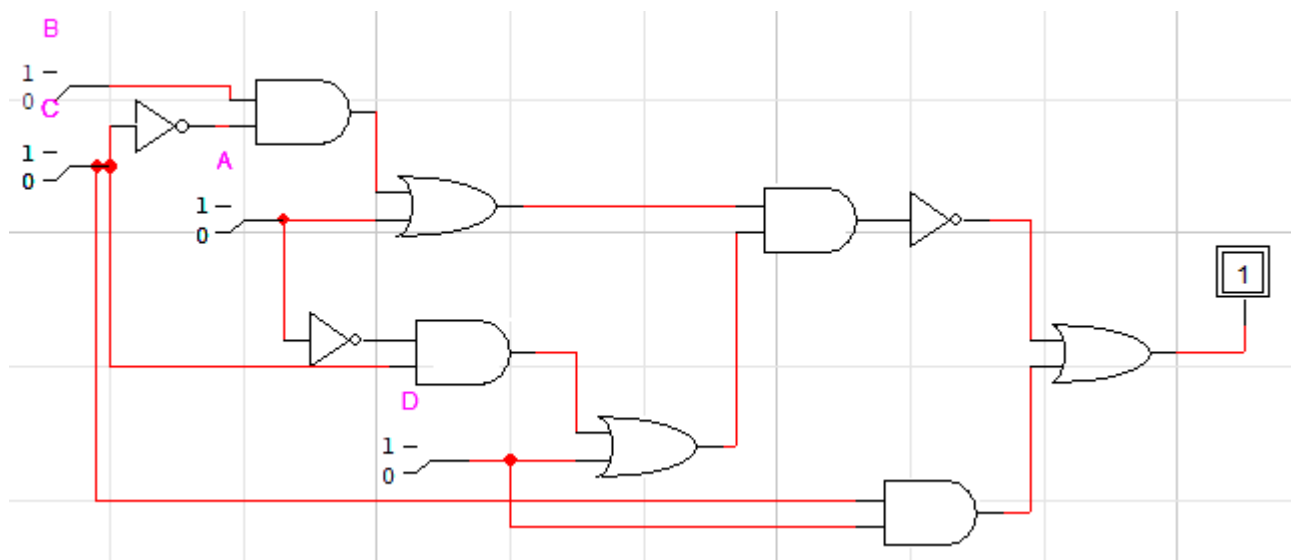
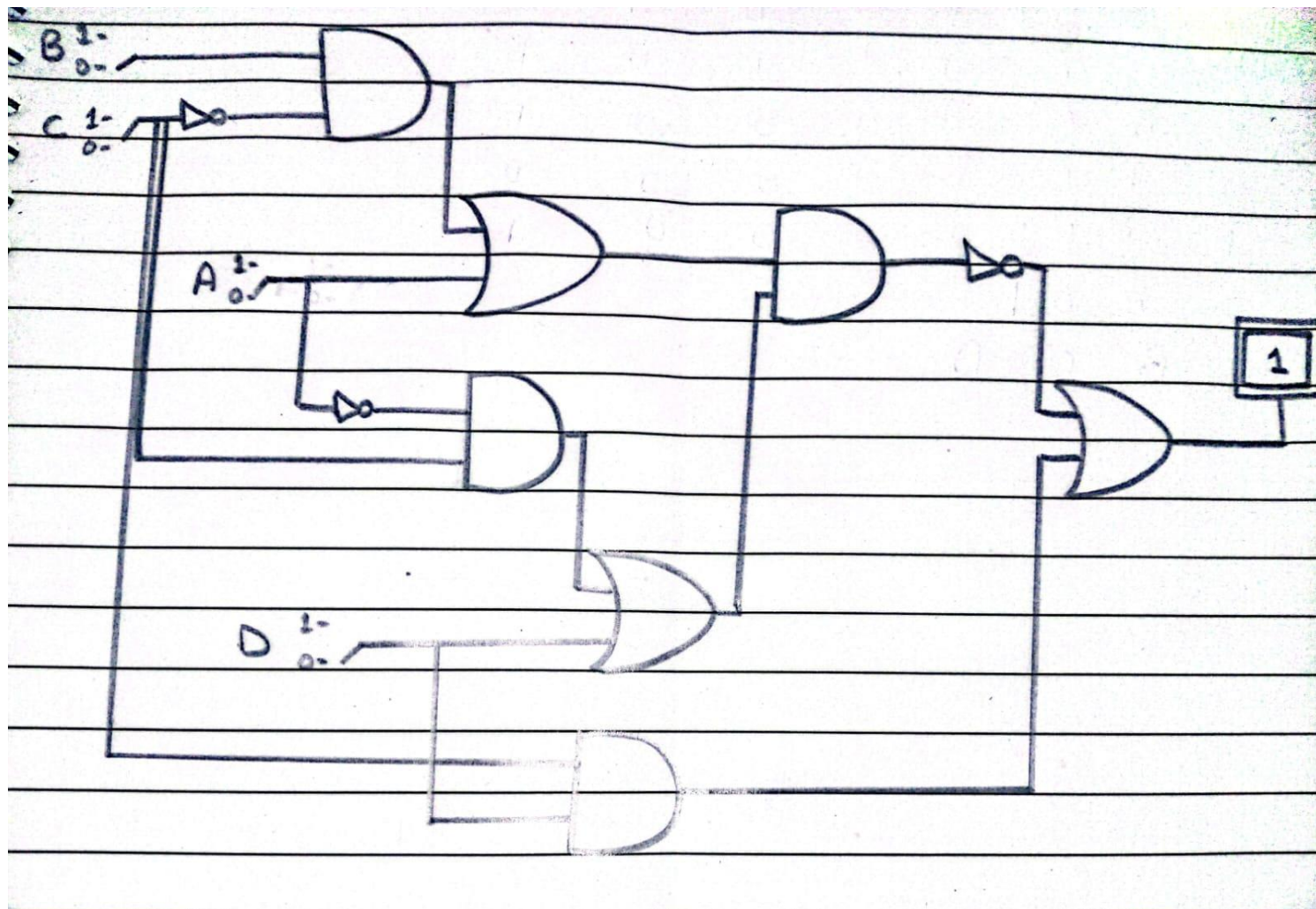


Task 1:

A	B	C	D	$\bar{A}$	$\bar{C}$	$B\bar{C}$	$\bar{A}\bar{C}$	$\overset{x=}{A+B\bar{C}}$	$\overset{y=}{D+\bar{A}\bar{C}}$	xy	$\overline{xy}$	CD	$\overline{xy}+CD$
0	0	0	0	1	1	0	1	0	1	0	1	0	1
0	0	0	1	1	1	0	1	0	1	0	1	0	1
0	0	1	0	1	0	0	0	0	0	0	1	0	1
0	1	0	0	1	1	1	1	1	1	1	0	0	0
1	0	0	0	0	1	0	0	1	0	0	1	0	1
1	1	0	0	0	1	1	0	1	0	0	1	0	1
1	0	1	0	0	0	0	0	1	0	0	1	0	1
1	0	0	1	0	1	0	0	1	1	1	0	0	0
0	1	1	0	1	0	0	0	0	0	0	1	0	1
0	1	0	1	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	0	0	0	0	1	0	1	1	1
1	1	1	0	0	0	0	0	1	0	0	1	0	1
1	1	0	1	0	1	1	0	1	1	1	0	0	0

1	0	1	1	0	0	0	0	1	1	1	0	1	1
0	1	1	1	1	0	0	0	0	1	0	1	1	1
1	1	1	1	0	0	0	0	1	1	1	0	1	1

Task 2:

$$(B \oplus C) \cdot A'$$

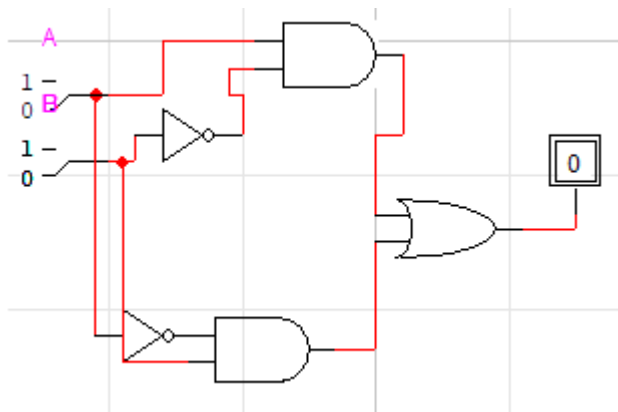
$$(\bar{B}C + B\bar{C}) \cdot A'$$

$$A'B'C + A'BC'$$

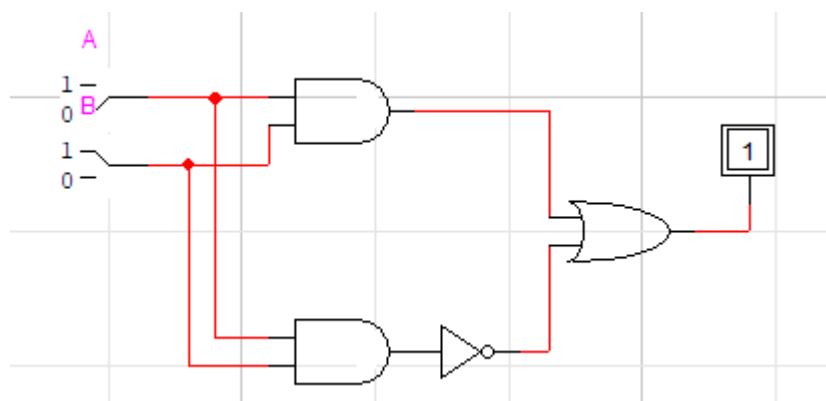
A	B	C	A'	B'	C'	A'B'C	A'BC'	A'B'C + A'BC'
0	0	0	1	1	1	0	0	0
1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	0	1	1
0	0	1	1	1	0	1	0	1
1	1	0	0	0	1	0	0	0
1	0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0
1	1	1	0	0	0	0	0	0

Task 3:

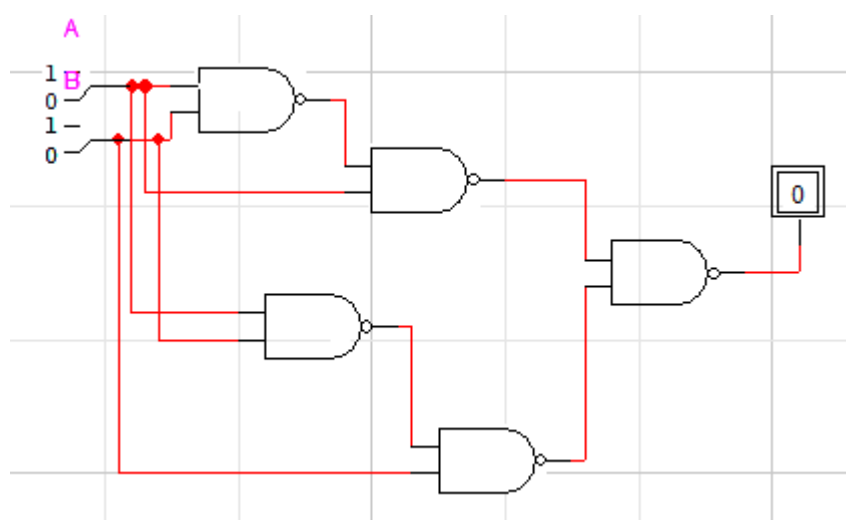
XOR:



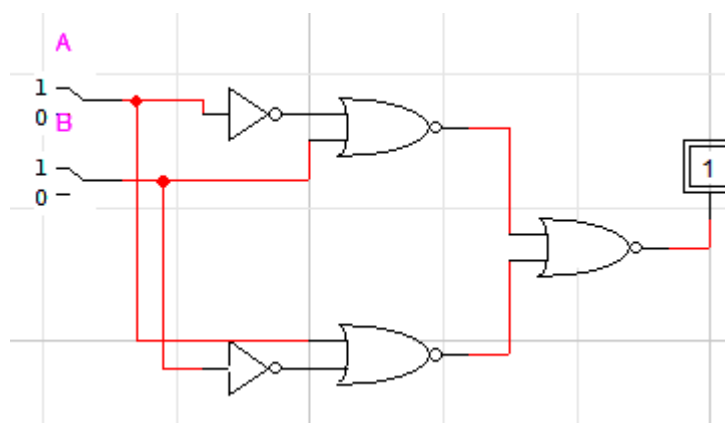
XNOR:

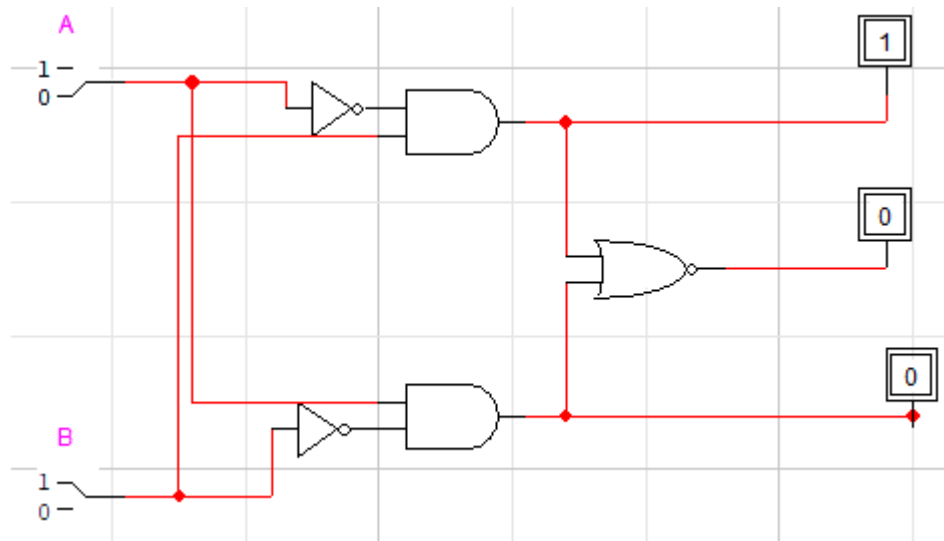
**Task 4:**

XOR:



XNOR:

**Task 5:**

**Task 6:**

$$AB + A(B+C) + B(B+C)$$

$$AB + AB + AC + BB + BC$$

$$AB + AC + BB + BC$$

$$AB + AC + B + BC$$

$$AB + B + BC + AC$$

$$B(A+1+C) + AC$$

$$B + AC$$

Task 7:

$$[A\bar{B}(C+BD)+\bar{A}\bar{B}]C$$

$$(A\bar{B}C+A\bar{B}BD+\bar{A}\bar{B})C$$

$$A\bar{B}C \cdot C + A\bar{B}BDC + \bar{A}\bar{B}C$$

$$A\bar{B}C + 0 + \bar{A}\bar{B}C$$

$$A\bar{B}C + \bar{A}\bar{B}C$$

$$\bar{B}C(A+\bar{A}) \Rightarrow \bar{B}C$$

Task 8:

$$\overline{AB + AC} + \bar{A}\bar{B}C$$

$$(\overline{A+B})(\overline{A+C}) + \bar{A}\bar{B}C$$

$$(\bar{A} + \bar{B})(\bar{A} + \bar{C}) + \bar{A}\bar{B}C$$

$$\bar{A} + \overline{BC} + \bar{A}\bar{B}C$$

$$\bar{A}(1 + \overline{BC}) + \overline{BC}$$

$$\bar{A} + \overline{BC}$$